TANAY RAGHUNANDAN SRINIVASA

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Education

MSc in Robotics - Technische Universiteit Delft 2028
B.Tech in Robotics & Autonomous Systems - Plaksha University, Mohali 2026

Work Experience

Project Trainee, UR Rao Satellite Center, Indian Space Research Organisation Jan 2026 - May 2026
- Developed Physics Informed Neural Networks to solve the **free-terminal time** optimal spacecraft attitude control problem.
- Solved the **Hamilton-Jacobi-Bellman** Partial Differential Equation, a necessary and sufficient condition for optimality, for a time-energy objective.
- Demonstrated **robust convergence** of the proposed controller architecture through Monte-Carlo simulations from various initial conditions.

Student Tutor, Plaksha University Aug 2025 - Dec 2025
- Conducted tutorials and office hours for FM206: Mechatronics, covering **sensors, actuators**, and the **differential drive robot model**.
- Designed and assisted students with the course project that included working with sensors such as encoders, IR, and the ultrasonic sensors.

Research Intern, UR Rao Satellite Center, Indian Space Research Organisation May 2025 - Jul 2025
- Developed Physics Informed Neural Networks to solve **Infinite Horizon Optimal Control Problems** for aerospace applications.
- Applied the **Extreme Theory of Functional Connections** to solve the Inverted Pendulum and Spacecraft Detumbling Optimal Control Problems.
- Derived the Hamilton-Jacobi-Bellman Equation, a necessary and sufficient condition for optimality, for a **constrained action space** and LQR cost.

Student Tutor, Plaksha University Feb 2025 - May 2025
- Conducted tutorials and computational labs for RO2001: System Dynamics and Control, covering **system modeling** and **performance analysis**.
- Designed and assisted students with the course balancing robot project, focusing on **motor characterization**, **sensor filtering**, and PID control.

Summer Research Intern, Robert Bosch Center for Cyber-Physical Systems, IISc May 2024 - Aug 2024
- Designed and Tuned a Linear Controller with a **Gain Scheduler** to Balance **BiSteering Two Wheeled Robot**, achieving balance for 76s.
- Researched and Measured Loaded and Unloaded **Motor Deadband** and Compensation techniques.
- Measured and Compared Settling Time and Overshoot of the **Bosch Sensor Fusion Algorithm** and a **Kalman Filter**.

Publications

Nanda, T.R., Shukla, A., Srinivasa, T.R., Bhargava, J., Chauhan, S. (2025). *Advancing Real-Time Crop Disease Detection on Edge Computing Devices Using Lightweight Convolutional Neural Networks*. In: Arai, K. (eds) Intelligent Systems and Applications. IntelliSys 2025. Lecture Notes in Networks and Systems, vol 1567. Springer, Cham. https://doi.org/10.1007/978-3-032-00071-2_33 .

T. R. Srinivasa and S. Kumar, *Solving Infinite-Horizon Optimal Control Problems using the Extreme Theory of Functional Connections*, 2025 Eleventh Indian Control Conference (ICC), Bangalore, India, 2025, pp. 147-152. https://doi.org/10.1109/ICC69100.2025.11372426

S. Kumar and T. R. Srinivasa, *HJB Optimal Feedback Guidance for Spacecraft Attitude Manuever via Physics-Informed Neural Networks*, AIAA SciTech Form 2027 (Accepted Aug 31, 2026).

Projects

Degradation Aware Optimal Battery Cycling for Smart Grids | *Prof. Vivek Deulkar* Aug 2025 - Jan 2026
- Developed RL agents to optimally cycle heterogeneous battery banks to fulfill demand-supply mismatches in **renewable energy generation**.
- Designed an **off-policy function-approximation** kernel inspired by the Extreme Learning Machine, outperforming handcrafted features by 2x.
- Researched and Integrated rainflow-based lithium ion depth-of-discharge **battery degradation** models to minimize operational costs.

SegwayRL | *Prof. Sandeep Manjanna* Jan 2025 - May 2025
- Designed and trained **PPO** and **DQN** agents to balance a segway robot, with training in both simulation and hardware.
- Developed a segway training and deployment environment using **OpenAI Gymnasium**, designed for 10ms sensing-actuation loop in hardware.
- Designed a **state-action reward function**, to penalize states far away from equilibrium, while penalizing large control efforts for small errors.

Low Altitude Remote Sensing (LARS) UAV for Crop Health Monitoring | *Prof. Sunita Chauhan* Jan 2024 - May 2025
- Developed a **ROS based communication** protocol to interface UAVs with ground robots to enable real-time autonomous crop monitoring.
- Developed a lightweight convolutional neural network based on MobileNet-v4 meant to be deployed on **constrained edge devices**.
- Calculated **Ground Sampling Distance (GSD)** at varying altitudes for UAVs, benchmarking imaging capabilities for precise crop health monitoring.

Relevant Coursework

AI4101: Machine Learning in Dynamic Environments | *Prof. Deepan Muthirayan* Grade: A
RO3003: Control Autonomy, Planning and Navigation | *Prof. Shashank Tamaskar* Grade: A
RO3002: Reinforcement Learning Fundamentals | *Prof. Sandeep Manjanna* Grade: A
RO2102: Dynamics and Simulation | *Prof. Andy Ruina* Grade: A
CP3001: Embedded Systems | *Prof. Srikant Srinivasan* Grade: A
AI3001: Deep Learning | *Prof. Anupam Sobti* Grade: A

Skills

Programming Languages: Python, MATLAB, C/C++, ROS2, Bash, LaTeX.
Micro-controllers: Raspberry Pi 3B+/4B/5, ESP-32, Arduino Uno/Nano, Teensy 4.1, STM32F103RB.

Positions of Responsibility

Mechanical Workshop Coorndinator, Robotics Lab, Plaksha University Aug 2024 - Jan 2025
Developed the floor plan, designed the workbench, procured tools, and set-up the mechanical workshop.

Achievements

Academic Achievement Award Jul 2026
Awarded for achieving the highest CGPA in the Robotics & Autonomous Systems Major in the graduating B.Tech Class of 2026.

Dean's List Nov 2025
Awarded for outstanding academic performance during the Academic Year 2024-2025.